

Text Representation Comparison Across Three Algorithms

LendingClub Default-Prediction Thesis — Lexicon vs TF-IDF vs FinBERT, Logistic Regression vs XGBoost vs Random Forest

Synthesized by `Dictionaries/scripts/08_three_tier_comparison_report.py` purely by reading the three tiers' already-written result files (Dictionaries/, TF-IDF/, FinBERT/) — no models were refit for this report.

Purpose

Each text-representation tier (Level 1 Lexicon, Level 2 TF-IDF, Level 3 FinBERT) was built and evaluated independently, by design (this repo's no-shared-module convention: each ablation script owns its own copy of the walk-forward fold logic, DeLong test, and bootstrap CI). This means no single existing document shows all three tiers against all three algorithms side by side. This report closes that gap by consolidating the already-computed results — it does not run any new model.

Important scope note: no combined arm (e.g. Structured+TF-IDF+FinBERT together) has been tested anywhere in this repo. Every row below adds exactly one text representation to the same Structured baseline. Stacking representations together is a distinct, untested question.

Structured-only reference (mean AUC, 14 folds)

Model	AUC
Logistic	0.6930 +/- 0.0152
XGBoost	0.6904 +/- 0.0191
RandomForest	0.6882 +/- 0.0199

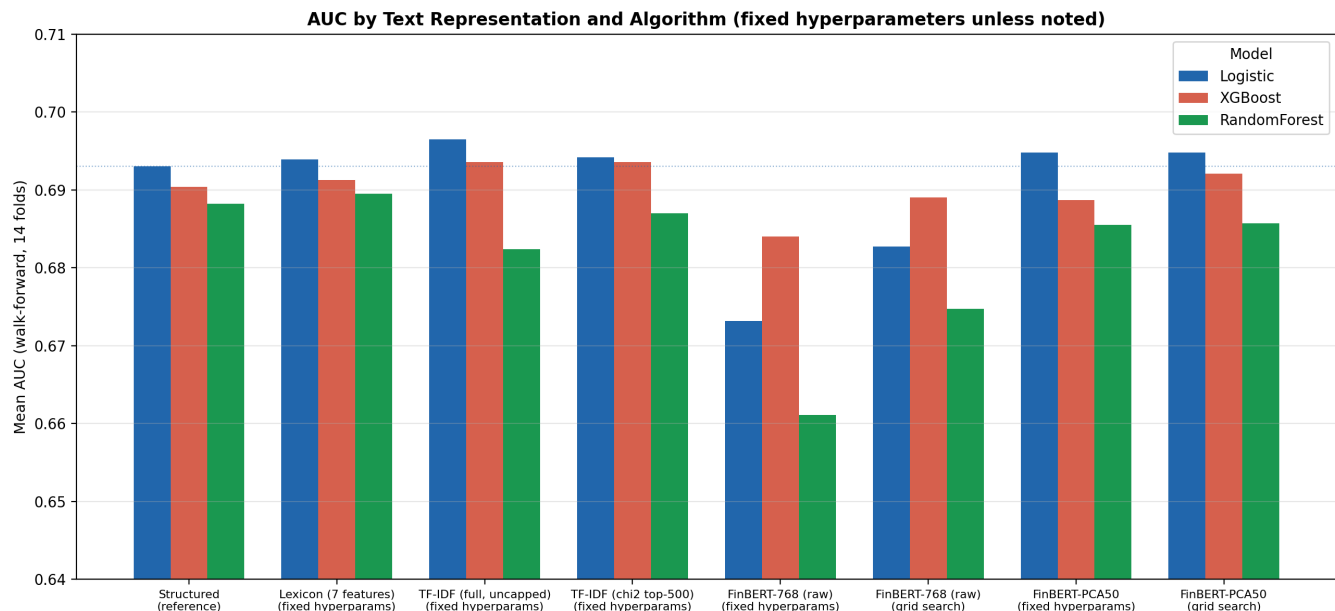
Master comparison table

Δ AUC is relative to the Structured-only reference above. "Significant folds" is the count (of 14) where a per-fold DeLong test found the difference significant at $p < 0.05$ (raw, not multiple-testing corrected). For FinBERT this comes from a dedicated script (`FinBERT/DELONG_VS_STRUCTURED.py`, added 2026-08-29) that refits Structured and both FinBERT arms side by side per fold, since the original baseline/gridsearch scripts only saved aggregate AUC, not paired per-fold predictions.

Representation	Config	Model	AUC	Delta AUC	Sig. folds
Lexicon (7 features)	fixed hyperparams	Logistic	0.6939	+0.0009	2/14
Lexicon (7 features)	fixed hyperparams	XGBoost	0.6913	+0.0009	0/14
Lexicon (7 features)	fixed hyperparams	RandomForest	0.6895	+0.0013	1/14
TF-IDF (full, uncapped)	fixed hyperparams	Logistic	0.6965	+0.0035	7/14
TF-IDF (full, uncapped)	fixed hyperparams	XGBoost	0.6936	+0.0032	3/14
TF-IDF (full, uncapped)	fixed hyperparams	RandomForest	0.6824	-0.0058	7/14
TF-IDF (chi2 top-500)	fixed hyperparams	Logistic	0.6942	+0.0012	2/14

TF-IDF (chi2 top-500)	fixed hyperparams	XGBoost	0.6936	+0.0032	2/14
TF-IDF (chi2 top-500)	fixed hyperparams	RandomForest	0.6870	-0.0012	3/14
FinBERT-768 (raw)	fixed hyperparams	Logistic	0.6732	-0.0198	5/14
FinBERT-768 (raw)	fixed hyperparams	XGBoost	0.6840	-0.0064	0/14
FinBERT-768 (raw)	fixed hyperparams	RandomForest	0.6611	-0.0271	12/14
FinBERT-768 (raw)	grid search	Logistic	0.6827	-0.0103	4/14
FinBERT-768 (raw)	grid search	XGBoost	0.6890	-0.0014	0/14
FinBERT-768 (raw)	grid search	RandomForest	0.6747	-0.0135	8/14
FinBERT-PCA50	fixed hyperparams	Logistic	0.6948	+0.0018	3/14
FinBERT-PCA50	fixed hyperparams	XGBoost	0.6887	-0.0017	1/14
FinBERT-PCA50	fixed hyperparams	RandomForest	0.6855	-0.0027	1/14
FinBERT-PCA50	grid search	Logistic	0.6948	+0.0018	3/14
FinBERT-PCA50	grid search	XGBoost	0.6921	+0.0017	1/14
FinBERT-PCA50	grid search	RandomForest	0.6857	-0.0025	1/14

AUC by representation and algorithm



Cross-cutting findings

- **RandomForest is structurally penalized by wide/dense text additions.** TF-IDF (full), FinBERT-768, and even FinBERT-PCA50 all reduce its AUC below the Structured-only reference; the only representation that helps it at all is the small 7-feature Lexicon (+0.0014). This matches FinBERT/RESULTS_COMPARISON.md §6a's direct finding: not one FinBERT dimension cracks RandomForest's global top-50 features, while Logistic and XGBoost both draw roughly half their top 50 from FinBERT — a model/algorithm mismatch given the fixed ``min_samples_leaf``, not a lack of signal in the text itself.
- **FinBERT-768 raw is the worst-performing representation tested, for every algorithm** (Δ AUC from -0.0064 to -0.0275 with fixed hyperparameters) — worse than adding nothing. It only becomes competitive after PCA-50 compression, and grid-search tuning recovers relatively little more on top of that. This is also the one case in the whole comparison with strong, consistent DeLong significance: RandomForest+FinBERT-768 loses to Structured-only in 12/14 folds at fixed hyperparameters (8/14 even after grid-search tuning) — not a borderline or noisy result, a reliably worse one.
- **Under fixed hyperparameters (the fairest apples-to-apples comparison, since this is how the Lexicon and TF-IDF arms were run), TF-IDF outperforms FinBERT-PCA50 for Logistic and XGBoost** (0.6948, 0.6936 vs 0.6887) — FinBERT only closes part of that gap with grid search, which TF-IDF was never given the chance to use here.
- **The Lexicon is the smallest and weakest representation in absolute terms, but the only one that never hurt any of the three algorithms.** Its effect sizes are consistently the smallest (+0.0009 to +0.0014) and, per Dictionaries/results/lexicon_ablation_report.md, rarely reach statistical significance (at most 2/14 folds) — consistent with the moderate Cohen's kappa (0.556) found during its own human validation.
- **Across all three tiers, effect sizes are small and mostly non-significant.** Even TF-IDF's best case (Logistic, full vocabulary) is only significant in 7 of 14 folds. The honest reading across this entire comparison is that

unstructured loan-description text adds, at best, a modest and algorithm-dependent increment over the 126 structured features already in the model — not a transformative one.

Data provenance / known asymmetries

The Structured-only reference AUC used throughout (0.6930 / 0.6904 / 0.6882) is taken from TF-IDF/results/ablation_summary.csv's own "Structured" arm, which is numerically identical to Dictionary's own Structured arm (both scripts fit it independently from the same STRUCT_COLS definition). FinBERT/RESULTS_COMPARISON.md reports a very slightly different Structured-only RandomForest figure (0.6886 vs 0.6882) from its own separate analysis/BASELINE.py run — a negligible (0.0004) cross-run numerical difference, not a methodological one, since all three reuse the identical STRUCT_COLS definition and walk-forward parameters.

The FinBERT vs Structured DeLong test (FinBERT/results/delong_vs_structured/, added 2026-08-29) refits both arms independently of the original baseline_768/gridsearch_768/baseline_pca50/gridsearch_pca50 runs, so its own AUC numbers can differ slightly (typically <0.002) from the AUC column above -- most likely from a scikit-learn version drift between when those scripts were first run and this later DeLong-only run (a 'penalty' deprecation warning appears in the newer run's logs that does not appear in the older ones). Only the Significant-folds count is taken from this test; the AUC/ Δ AUC columns still come from each arm's own original run.